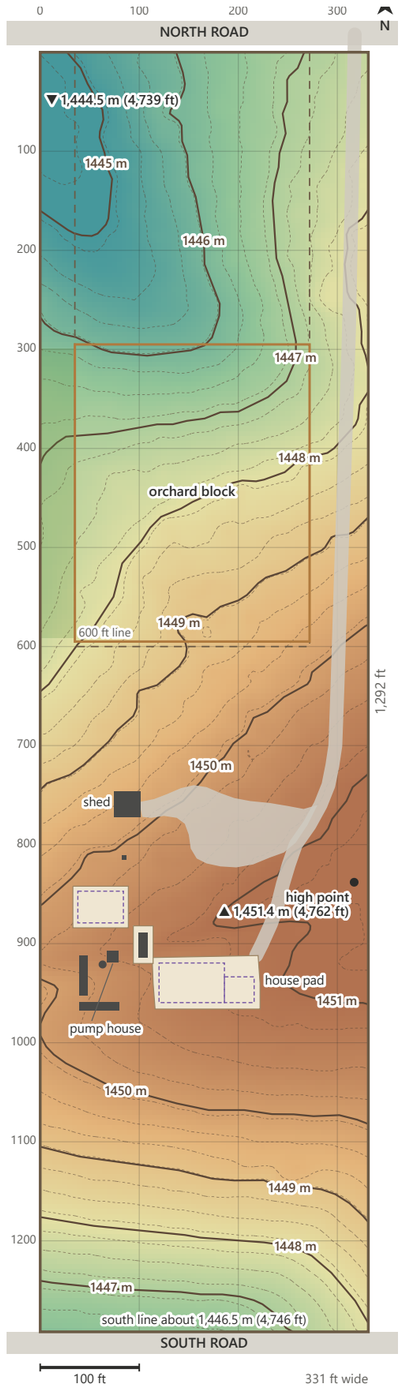
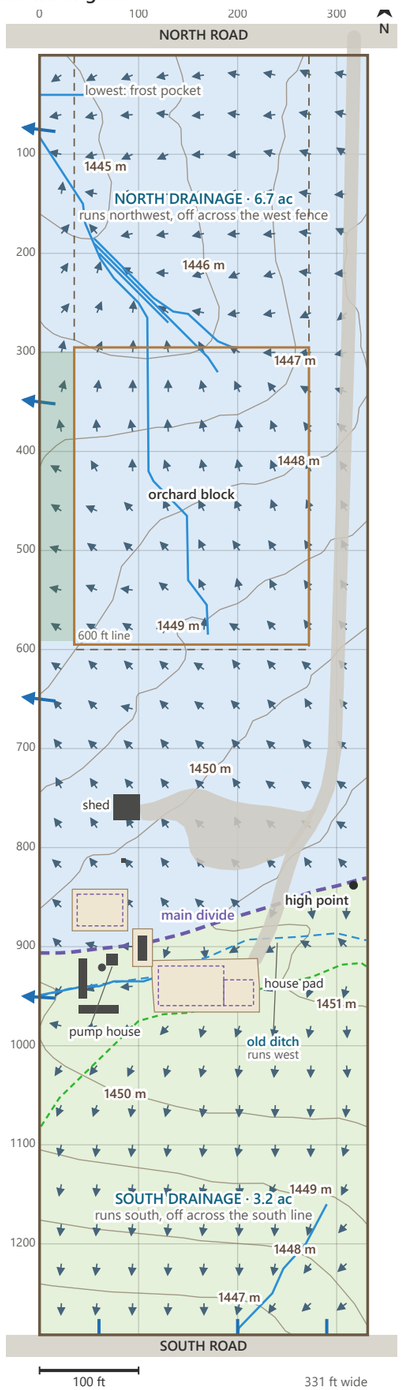
Mountain Prairie Acres: Whole-Site Terrain and Water

The whole 10-acre lot to scale, north up, as it lies today: elevation and contours, where water goes, and slope. Base maps for planning future projects anywhere on the property.

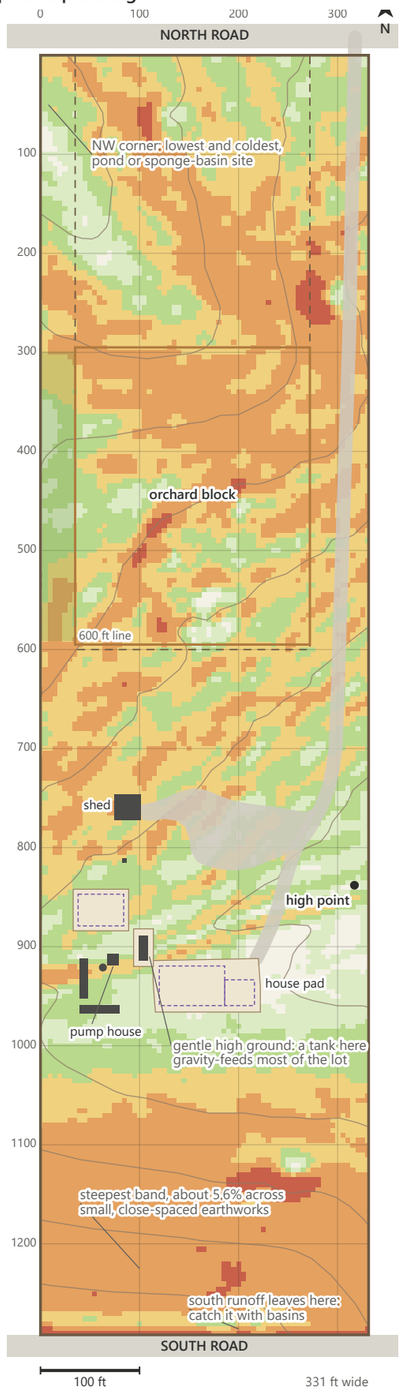
1 · Elevation and contours



2 · Where water goes



3 · Slope and planning



LOT **9.8 acres** 331 × 1,292 ft, fence to fence	HIGHEST **1,451.4 m** about 4,762 ft, east fence beside the homestead
LOWEST **1,444.5 m** about 4,739 ft, west fence near the NW corner	RELIEF **about 22 ft** from the high point down to the NW corner
NORTH DRAINAGE **6.7 acres** runs northwest, off across the west fence	SOUTH DRAINAGE **3.2 acres** 2.5 ac across the south line, 0.7 ac west by the old ditch
TYPICAL SLOPE **3.6%** most ground 1.9%–5.2%	SOIL **Simonton loam** loam over clay loam; deep, well drained

Legend

Map 1 elevation, m (4,741–4,760 ft)

1445 1446 1447 1448 1449 1450 1451

- 1 m contour (lidar, labeled in meters)
- 1 ft contour (lidar)
- Flow path: runoff from a quarter acre or more gathers
- Split line: south of it, runoff leaves across the south line
- Main divide: north vs. south drainage
- Old ditch (lidar, before the homestead pads)
- Downhill direction
- North drainage area
- South drainage area

Map 3 slope: under 1% · 1–2% · 2–3% · 3–4% · 4–6% · over 6%

- Fence (lot line)
- Driveway and road-mix yard
- Orchard block R1–R15 (inside the dashed 600 ft strip)
- Shed, containers, pump house, tank
- Graded pad (house, workshop, RV)
- Planned house, garage, and workshop

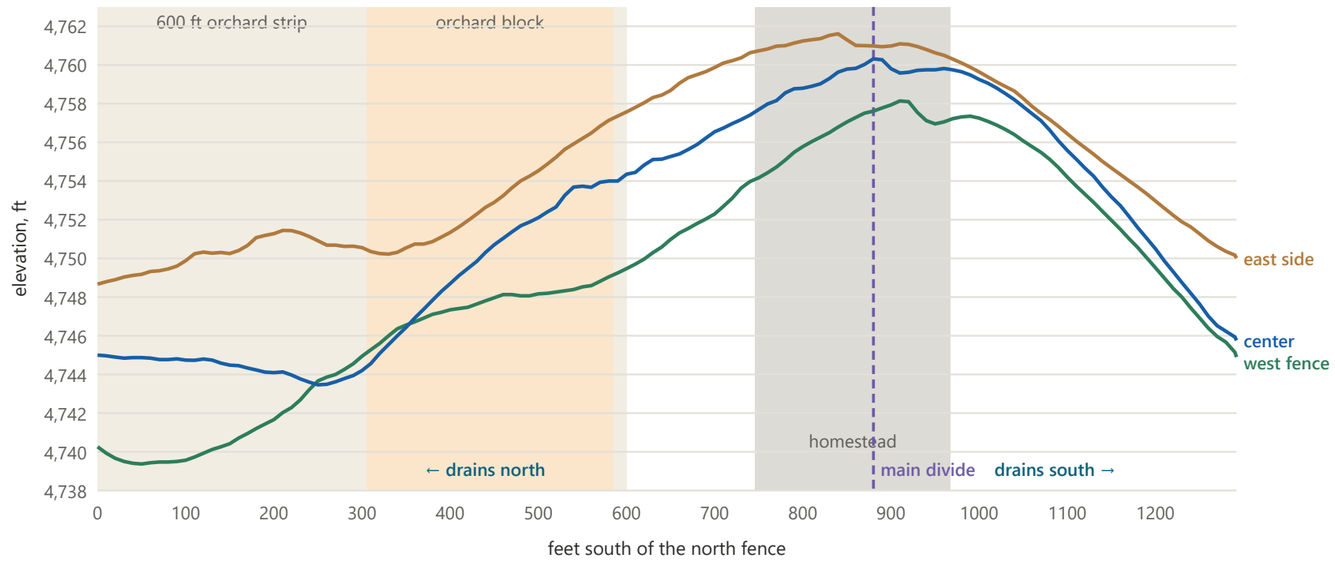
For future projects

- High ground: the homestead and the east fence beside it (about 1,451.4 m) top both drainages. The 3,000 gal tank by the pump house stands on ground about 19 ft above the low NW corner: roughly 8 psi by gravity there, less closer in.
- North drainage (6.7 acres): everything north of the main divide runs northwest and leaves across the west fence (6.7 acres; only a sliver crosses the north fence). The main runoff line runs from the middle of the orchard block to the west fence about 75 ft south of the NW corner. That corner is the lowest, wettest, and coldest spot: a place for a pond, sponge basin, or wet-tolerant plants, not frost-tender ones.
- South drainage (3.2 acres): 2.5 acres leave across the south line; the strip just south of the divide (0.7 acre) runs west along the old ditch past the house pad and containers. Basins or swales keep that water on your ground.
- Blue flow paths mark where runoff gathers. Keep buildings, driveways, and tanks out of them; use them for swales, basins, or a pond.
- Slope: most of the lot is gentle (typical 3.6%). Flatter ground suits buildings, gardens, and level basins; on the steeper bands, use smaller, closer-spaced earthworks and keep the ground covered.
- Cold air drains like water, so frost settles first in the NW corner and along the low west side.
- To place a project, read its coordinates off the 100 ft grid (or a 50 ft cell on the Site Master Grid page), then its elevation from map 1 or the spot table on page 2. Check with a level before building.

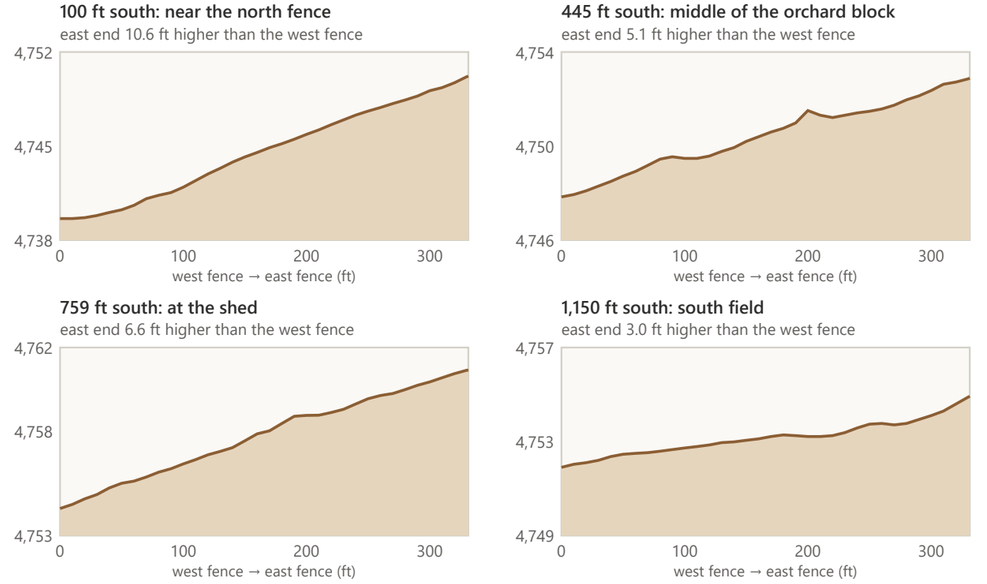
All three maps share one scale, north up. Coordinates are master-grid lot feet: east of the west fence and south of the north fence; grid lines are 100 ft apart (50 ft cells are on the Site Master Grid). Elevations from USGS lidar, good to a few tenths of a foot on open ground.

Profiles, Spot Elevations, and Key Points

North–south profiles, north fence to south fence



East–west cross-sections (vertical scale exaggerated)



Spot elevations in feet, every 100 ft south of the north fence

South ↓ / east →	0	50	100	150	200	250	300	331
0	4,740	4,742	4,743	4,744	4,746	4,747	4,748	4,749
100	4,740	4,740	4,742	4,744	4,746	4,748	4,749	4,750
200	4,742	4,741	4,742	4,743	4,746	4,748	4,750	4,751
300	4,745	4,744	4,744	4,744	4,745	4,747	4,749	4,751
400	4,747	4,748	4,748	4,749	4,749	4,750	4,751	4,752
500	4,748	4,749	4,751	4,752	4,753	4,753	4,754	4,755
600	4,749	4,751	4,753	4,754	4,755	4,756	4,757	4,758
700	4,752	4,753	4,755	4,756	4,758	4,758	4,759	4,760
800	4,756	4,757	4,758	4,759	4,759	4,760	4,761	4,761
900	4,758	4,759	4,759	4,760	4,760	4,760	4,761	4,761
1,000	4,757	4,758	4,759	4,759	4,759	4,760	4,760	4,760
1,100	4,754	4,755	4,755	4,755	4,756	4,756	4,756	4,757
1,200	4,749	4,750	4,750	4,750	4,751	4,751	4,752	4,753
1,292	4,745	4,745	4,746	4,746	4,746	4,747	4,749	4,750

Columns are feet east of the west fence. Shading follows the elevation colors on map 1. One meter is 3.28 ft.

Key points

Point	x, y (ft)	m	ft	Above NW corner
NW corner	0, 0	1,444.8	4,740	0.0 ft
Low point, west fence	0, 50	1,444.5	4,739	-0.9 ft
NE corner	331, 0	1,447.4	4,749	8.7 ft
R15 row, center	154, 305	1,446.1	4,744	4.2 ft
R1 row, center	154, 585	1,449.1	4,754	14.2 ft
Shed	88, 759	1,449.7	4,756	16.1 ft
Pump house and tank	73, 913	1,450.6	4,759	19.0 ft
House pad, center	168, 939	1,450.8	4,760	19.7 ft
High point, east fence	317, 838	1,451.3	4,762	21.5 ft
Main divide, center	166, 880	1,450.9	4,760	20.2 ft
SW corner	0, 1,292	1,446.2	4,745	4.8 ft
SE corner	331, 1,292	1,447.9	4,750	10.2 ft

How these maps were made

- Elevations: USGS 3D Elevation Program (3DEP) 1 m lidar, fitted to the fence lines on USGS NAIP aerial imagery (331 × 1,292 ft fence to fence) and smoothed to about 5 ft. 1 m and 1 ft contours, slope, flow arrows, profiles, and spot heights come from that surface (data/site_terrain.js). The lidar was flown before the homestead pads were graded.
- Flow paths, the main divide, the split line, and the drainage areas come from routing water downhill across that surface (each 5 ft cell to its steepest neighbor).
- Homestead, driveway, and yard from the owner's drone photos (October 2026), fitted to the same imagery and lidar; planned buildings where the owner described them (data/site_grid.js). Orchard layout from the pollination grid (October 4, 2026), with R1 about 15 ft inside the south end of the 600 ft strip.
- An earlier version of these maps traced the owner's USGS 1 m map screenshots; the screenshots were tilted, which shifted features by 10–30 ft, so the maps now use the lidar directly.
- Soil: USDA NRCS Web Soil Survey report for the property, map unit 149, Simonton loam, 2 to 12 percent slopes.